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Model-Based Parameters Estimation and Design Optimization for Electric Race Vehicles

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Abstract

In the realm of electric vehicle design and development, critical vehicle parameters are estimated by coast down test. However, other drag factors can introduce errors to the estimation, including inertial and friction of rotating parts. Additionally, certain parameters are challenging to measure directly, such as drag coefficient, driveline losses and the frontal area of vehicles like go-karts. To address this challenge, a method of model-based parameter estimation is

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proposed in this research. An electric vehicle and powertrain model of a go-kart are developed, incorporating uncertain parameters like rolling resistance coefficient and air density. The target vehicle undergoes testing, with vehicle speed and motor torque recorded concurrently. Subsequently, the MATLAB/Simulink parameter estimation tool is applied to fine-tune the uncertain parameters, aligning the simulated vehicle speed and torque with the recorded test data. The optimization reduces speed and torque error between simulation values and test data by 24.76% and 15.58% respectively comparing to coast down model. Once optimized, the model is validated through simulation using another set of recorded drive cycle data reduces speed and torque error by 48.46% and 30.63% respectively comparing to coast down model. With a validated model boasting high confidence, design optimization tools are utilized to enhance the traction motor's pulley ratio. The optimized pulley ratio leads to a 33.80% improvement in energy efficiency, thereby enhancing the overall performance of the electric go-kart.

1.Introduction

Vehicle modeling is an essential tool in the automotive industry, serving as a critical component in vehicle design, control development, validation, and calibration. By simulating vehicle dynamics and systems, vehicle performance under various conditions can be predicted without the need for physical prototypes. This not only speeds up the development process but also reduces costs and allows for a more extensive testing range than would be feasible with physical models alone (Adegbohun et al. 2021, Montaleza et al. 2022, Genikomasakis and Mitrentsis 2017). Furthermore, vehicle modeling facilitates the refinement of control systems, ensuring that they operate efficiently and reliably in real-world scenarios, thereby enhancing vehicle safety and performance (Adegbohun et al. 2021, Enang and Bannister 2017, Krasopoulus et al. 2017). One effective method for modeling a vehicle and extracting its parameters is through a coast-down test. This type of test involves allowing a vehicle to coast freely from a set speed until it comes to a stop, all the while measuring various parameters such as speed, acceleration, and distance.

By analyzing the deceleration of the vehicle, important parameters like aerodynamic drag, rolling resistance, and mechanical losses within the drivetrain can be estimated (Adegbohun et al. 2021, Baghdadi et al. 2013, Chehresaz 2013, Genikomasakis and Mitrentsis 2017, Yoong et al. 2010). However, the coast-down test lacks in actuating the driver's behaviors as well as dynamic interactions as it begins with the predetermined parameters resulting oversimplification and reduced accuracy while trying to model the powertrain under varying drive cycles. In addition, there might be inaccuracies while estimating the vehicle parameters because of the inertial drag and rotational friction. Another challenge is measuring specific parameters such as the frontal area can be challenging for certain vehicles like golf carts and go-karts. A study on determining the vehicle parameters using coast-down testing for on-road bicycles (Tengattini, S. and Bigazzi, A. 2018) highlights that the accuracy of such tests is reduced due to factors like inconsistent wind, slope, and surface conditions. Even though the coast-down tests are expected to generate realistic parameter estimation for the modeling of on-road bicyclists, the precision of the test was lower and estimated to have higher margin of error with a rolling resistance coefficient of 0.0064 ± 0.0013 and effective frontal area of 0.63 ± 0.11 m^2.

To overcome these challenges, a model-based approach is proposed for estimating the parameters of electric vehicles, coupled with an optimization technique to minimize error and enhance accuracy. This method utilizes a forward vehicle model, initially assuming multiple uncertain parameters. These parameters are automatically estimated using a parameter estimation tool to align the simulated speed and torque with actual test data. Once the discrepancies in speed and torque are reduced to negligible levels, the vehicle model is considered to be accurately estimated and ready for further design. Unlike traditional coast-down vehicle modeling, which primarily focuses on vehicle speed, the proposed method also considers traction torque. This dual validation approach significantly improves the model's accuracy.

2.Modeling and optimization method

In this research, the architecture of the forward vehicle model is depicted in Figure 1. The model begins with a driver block that receives a reference speed profile. This profile is compared to the vehicle speed simulated from the vehicle dynamics output to identify any discrepancies between the actual and desired speeds. The resulting difference is input into a Proportional-Integral (PI) controller, which processes this data to generate signals for throttle and brake positions. These positions subsequently determine the torque request within the electric powertrain, ensuring the vehicle responds appropriately to achieve the desired speed.

The electric powertrain block incorporates a torque map for the electric motor, which is based on datasheet from supplier and depicted in Figure 2. The actual torque generated by the motor is determined by the motor speed and the

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position of the throttle pedal. This setup ensures that the powertrain's output accurately reflects the driver's input and the motor's operating conditions.

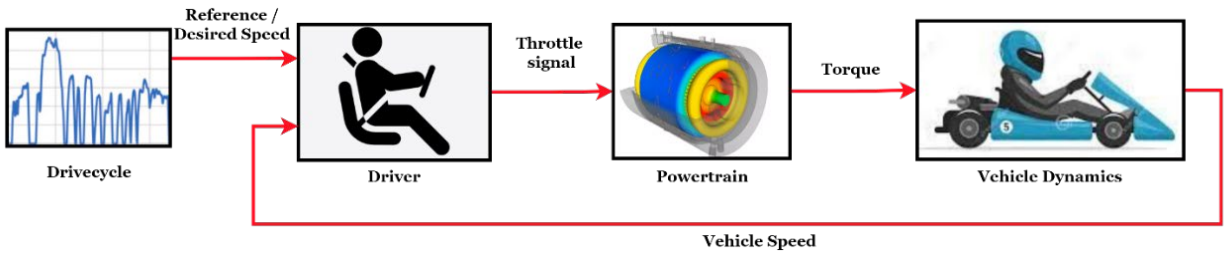


Figure 1. Forward Vehicle Model Architecture

The vehicle dynamics block forms the core component of the simulation, focusing on the road load forces acting on the vehicle. This block includes rolling resistance, aerodynamic drag force, and hill climbing force (Adegbohun et al. 2021, Chehresaz 2013). The total road load force is calculated by equation (1)

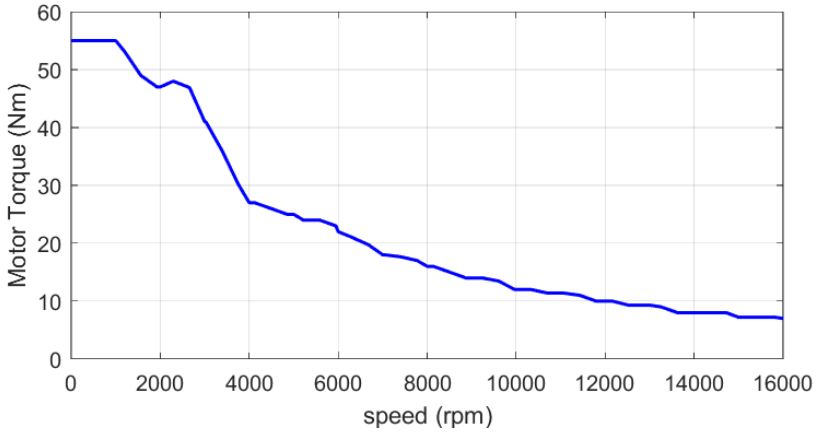


Figure 2. Motor Torque VS. Speed Map

$$F_{RL} = F_{RR} + F_{AD} + F_{HC} \tag{1}$$

Where F_{RR} is the tire rolling resistance, F_{AD} is the air drag force, and F_{HC} is the hill climbing resistance.

The rolling resistance F_{RR} can be expressed as:

$$F_{RR} = m \cdot g \cdot f_R \cdot \cos\theta \tag{2}$$

Where m is the total weight of the vehicle including driver and payload, g is the gravitational force (9.8 N/kg), f_R is the tire rolling friction coefficient, and θ is the road grade in degrees.

Air drag force F_{AD} is expressed as:

$$F_{AD} = 0.5 \cdot \rho_{air} \cdot A \cdot C_d \cdot V^2 \tag{3}$$

Where ρ_{air} is the air density, A is the vehicle frontal cross-sectional area, C_d is the air drag coefficient and V is the vehicle speed.

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Hill climbing resistance F_{HC} is calculated as:

$$F_{HC} = m \cdot g \cdot \sin\theta \tag{4}$$

The vehicle speed is calculated by equation (5) then fed back to the driver block.

$$V = \int_0^t \frac{F_{TR} - F_{RL} - F_{BR}}{K_m \cdot m} \tag{5}$$

Where F_{TR} is the traction force calculated by equation (6), F_{BR} is the break force, K_m is the mass factor which accounts for the rotational energy in the powertrain.

$$F_{TR} = \frac{T_M \cdot G_r}{r_t} \tag{6}$$

Where T_M is the motor torque, G_r is the pulley ratio and r_t is the tire radius.

The forward electric vehicle model developed in this study is based on an in-house designed go-kart, as illustrated in Figure 3. The go-kart's powertrain operates with a 48V traction motor coupled with a pulley transmission system. Known parameters such as the vehicle weight, driver weight, and pulley ratio are integrated into the model to ensure its accuracy. Additionally, other uncertain vehicle parameters, along with control coefficients, are initially assumed and listed in Table 1. These initial values are subject to estimation based on optimization and testing to refine the model's accuracy and reliability. The modeled vehicle underwent testing through two separate random drive cycles on a race track. The initial drive cycle (drive cycle 1) was employed for the estimation of uncertain parameters, while the subsequent cycle (drive cycle 2) served as a validation phase without any tuning of parameters. The driving cycles are shown in Figures 4 and 6, respectively.



Figure 3. Electric go-kart used for modeling & Optimization

During the testing phase, vehicle speed and motor torque data are collected and transmitted via the Controller Area Network (CAN). This data is then recorded with TSMaster CAN-USB hardware and interface at 10ms frequency systematically for analysis. Figure 4 displays a comparison between the simulated vehicle speed and torque—before estimation—and the test data from the first drive cycle. Coast down test has been performed as well to estimate vehicle parameters as baseline.

It is evident from Figure 4 and 5 that there is a noticeable discrepancy between the initial coast-down model and actual test data. A quantified analysis reveals that the average errors for speed and torque are 0.3368 m/s and 2.721 N/m respectively. These results indicate that the initial assumed values of the uncertain parameters are inadequate for accurately modeling the target vehicle which can be visualized in Figure 4 and 5.

The MATLAB/Simulink Parameter Estimation toolbox is employed to estimate the parameters outlined in Table 1. Through the application of least square optimization (Parameter Estimation MathWorks, Van et al. 2005), the parameters are adjusted within a reasonable range to minimize the discrepancies in speed and torque between the simulation and actual test data. The results of these estimations are depicted in Figure 4 and 5, showing the vehicle speed and torque simulation data with the newly estimated parameters. This adjustment significantly enhances the

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model's accuracy from coast down baseline, reducing the average speed error by 24.76% from 0.3368 m/s to 0.2534 m/s and the average torque error by 15.58% from 2.721 Nm to 2.297 Nm.

Table 1. Uncertain vehicle and control parameters initial assumptions and estimated values

Description	Representation	Initial Values	Estimated values
Air Drag Coefficient	C_d	0.142	0.654
Front Area	A	1.65 m^2	1.019
Proportional Gain	P	10	0.00752
Integral Gain	I	0.1	0.00053
Mass factor	K_m	1	1.13
Rolling Resistance	f_R	0.0068	0.0563

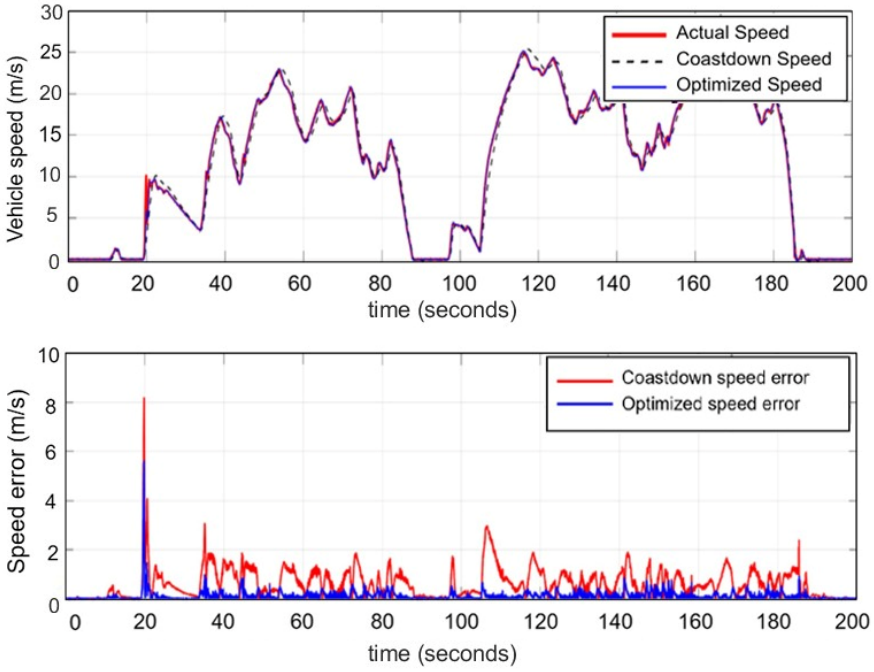


Figure 4. Actual / estimated speeds and error for drive cycle 1

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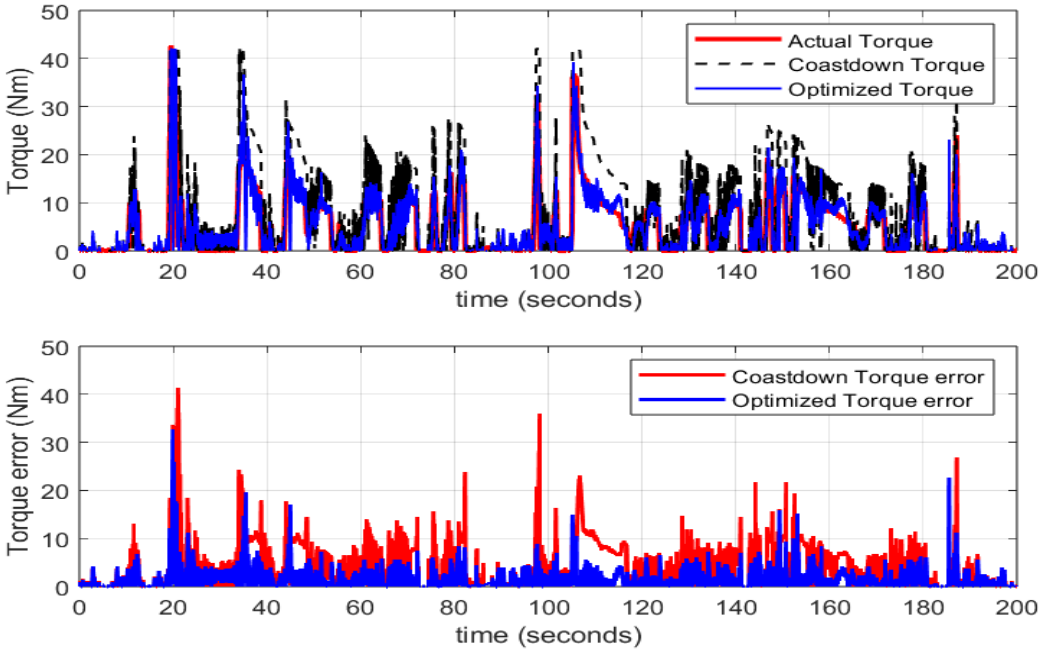


Figure 5. Actual / estimated torques and error for drive cycle 1

Drive cycle 2 is utilized to validate the results of the model-based parameters estimation. The comparisons of the simulated vehicle speed and torque, both before and after estimation, are illustrated in Figure 6 and Figure 7 respectively. A quantified analysis of these figures shows that the average speed and torque errors have been reduced from 0.2175 m/s to 0.1121 m/s by 48.46%, and from 2.951 Nm to 2.047 Nm by 30.63% compared to coast down model, respectively. This significant reduction in errors confirms that the model with parameters estimated achieves a high level of accuracy and is robust enough to handle various types of drive cycles.

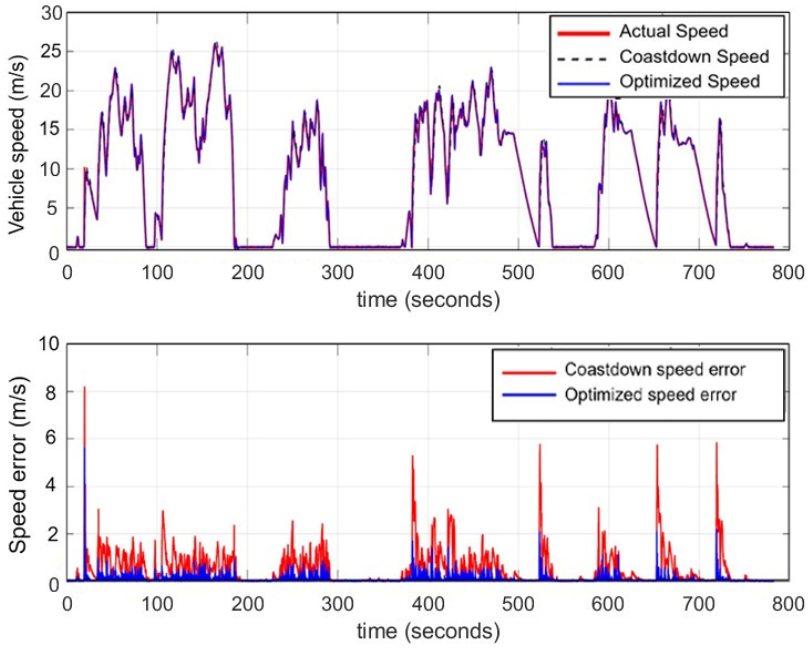


Figure 6. Actual / estimated speed and error for drive cycle 2

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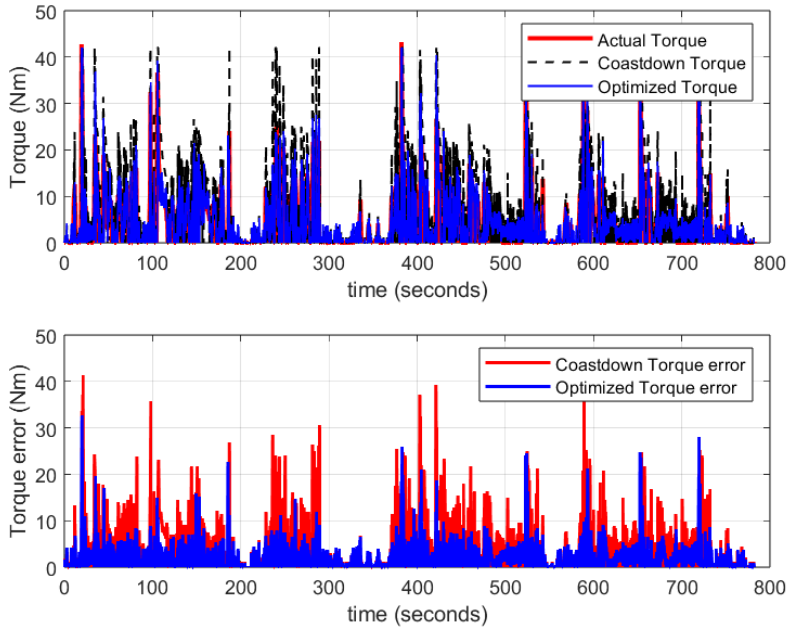


Figure 7. Actual / estimated torque and error for drive cycle 2

3. Model-Based Powertrain Design Optimization

With the vehicle model parameters accurately estimated and validated, it now serves as a dependable tool for model-based design, aimed at enhancing performance. The primary goal in this research is to minimize energy consumption by optimizing the pulley ratio within the powertrain. To facilitate this, a motor torque map has been integrated into the model. The MATLAB/Simulink Response Optimization toolbox (response Optimization, MathWorks) is employed to fine-tune the pulley ratio, ensuring optimal performance. Drive cycles 1 and 2 are simulated consecutively to refine the design. The pulley ratio adjustment, from 5.625 to 3.645, successfully maintains the same speed and torque profile, as demonstrated in Figure 8.

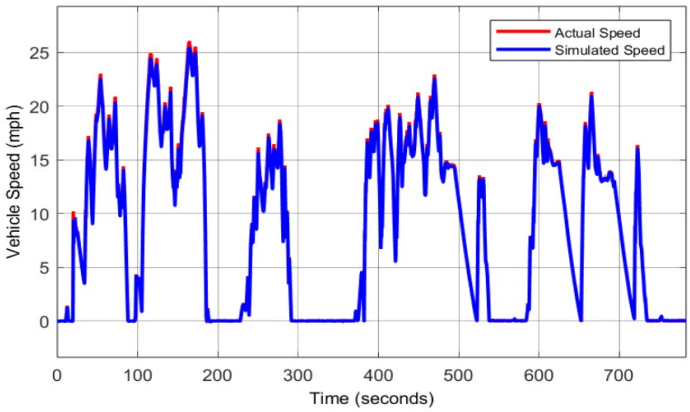


Figure 8. Vehicle speed of combined drive cycle with pulley ratio optimization

This adjustment confirms that the optimized pulley ratio provides sufficient torque to achieve the desired speed. Moreover, simulations indicate that the optimized pulley ratio leads to a significant reduction in energy consumption, decreasing from 0.51 kWh to 0.33 kWh for the combined drive cycles, which translates to a reduction of 33.8%. This outcome underscores the effectiveness of the model-based design approach in achieving more efficient vehicle

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operation. The motor operating points and efficiency Gaussian distribution for the combined drive cycle, both before and after the design optimization, are displayed in Figure 9 and 10 respectively.

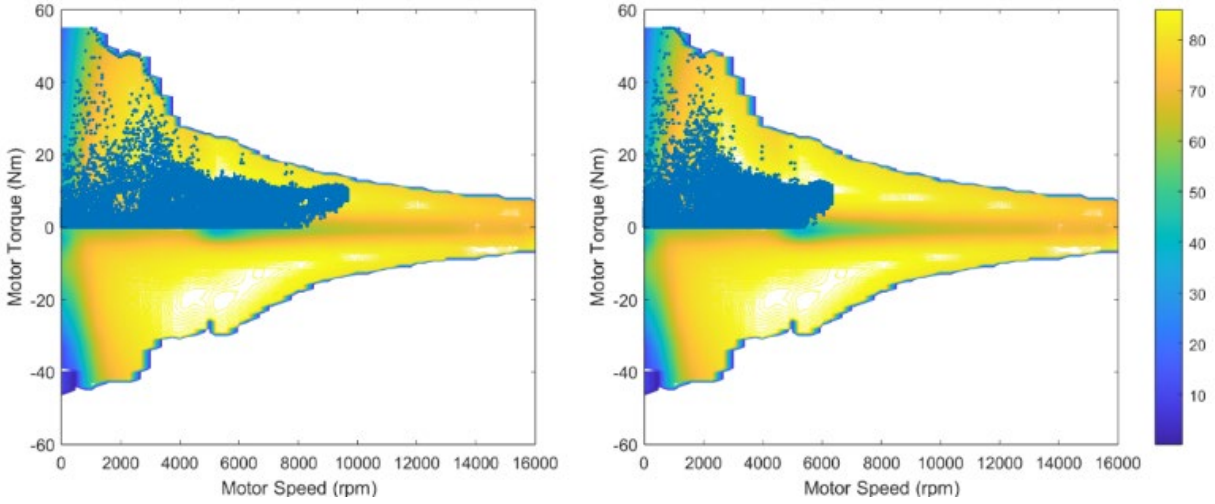


Figure 9. Motor operating points (i) before and (ii) after pulley ratio optimization

In Figure 10, the probability density function f(x) is calculated by equation 7.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \tag{7}$$

where $f(x)$ is the probability density function, x is efficiency in this case, μ is the mean of x and σ is the standard deviation of x. Both the operating points map and Gaussian distribution of efficiency validate that the optimized pulley ratio does indeed help the motor operate more frequently in higher efficiency areas.

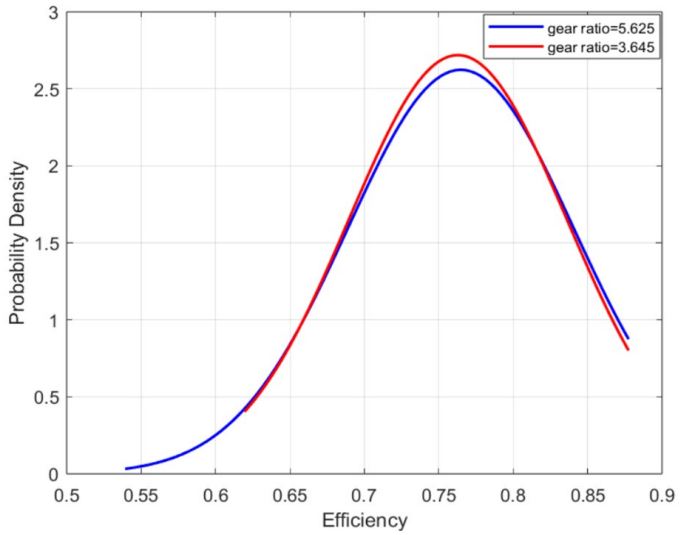


Figure 10. Gaussian Distribution of Motor Efficiency

It is essential to note that both acceleration performance and energy efficiency are critical factors in vehicle design, especially under the guidelines of the evGrandPrix competition. To evaluate the impact of pulley ratio optimization on acceleration, simulations were conducted to measure 0-60 mph acceleration times at full throttle. The results for different pulley ratios are depicted in Figure 11, and the incremental times to reach 10, 20, 30...60 mph for each ratio

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are detailed in Table 2. The simulation data reveals that when the pulley ratio is altered from 5.625 to 3.645, the vehicle still maintains satisfactory acceleration performance. Thus, the optimized pulley ratio of 3.645 will be adopted for the next vehicle iteration to enhance energy efficiency without compromising acceleration capabilities. This balance is crucial for meeting the competitive standards set by evGrandPrix, ensuring the vehicle remains competitive in both performance metrics.

Table 2. Comparison of acceleration performance for 0–60 mph with two pulley ratios

Acceleration	Pulley ratio=5.625 Time (seconds)	Pulley ratio=3.645 Time (seconds)	Time difference (seconds)
0–10 mph	0.1679	0.2517	0.0838
0–20 mph	0.3723	0.5354	0.1631
0–30 mph	0.7206	0.8705	0.1499
0–40 mph	1.1834	1.4211	0.2377
0–50 mph	1.9004	2.1758	0.2754
0–60 mph	3.3169	3.4117	0.0948

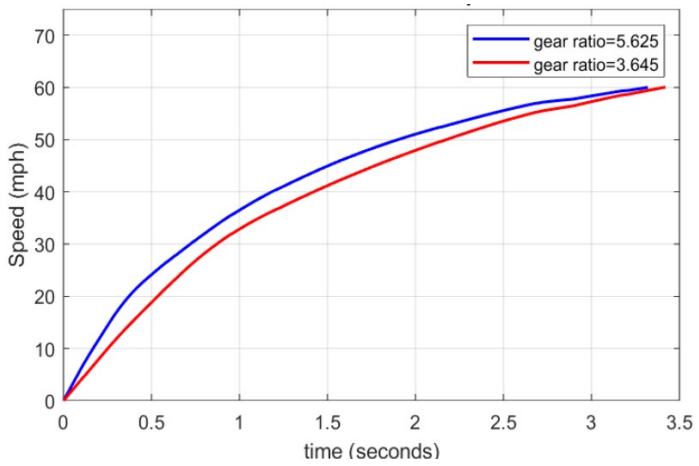


Figure 11. 0-60 mph Acceleration of 2 Pulley Ratios

4. Conclusions

This paper introduces a model-based method for estimating vehicle parameters in electric vehicles. A forward vehicle model was developed, initially assuming uncertain parameters. The MATLAB/Simulink parameter estimation toolbox was employed to automatically refine these parameters based on empirical test data. Both vehicle speed and motor torque were targeted for parameter estimation to minimize discrepancies between simulated outcomes and test data from drive cycle 1. This estimation process significantly reduced the average speed error by 24.76% from 0.3368 m/s to 0.2534 m/s and the average torque error by 15.58% from 2.721 Nm to 2.297 Nm comparing to coast down model. Drive cycle 2 was then comparing to coast down model utilized to validate the effectiveness of the model-based parameter estimation. The results showed substantial improvements, with average speed and torque errors decreasing by from 0.2175 m/s to 0.1121 m/s by 48.46%, and from 2.951 Nm to 2.047 Nm by 30.63% comparing to coast down model, respectively.

Following the validation, a pulley ratio design optimization was conducted to maximize energy efficiency. By simulating drive cycles 1 and 2 back-to-back, the pulley ratio was adjusted from 5.625 to 3.645. This adjustment contributed to a reduction in energy consumption for the combined drive cycle by 33.8%, lowering it from 0.51 kWh to 0.33 kWh.

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Biographies

Sujan Thapa is graduate student in the Electrical and Computer Engineering Department of the College of Engineering at Kettering University and currently pursuing MSE in Mobility Systems with a background in Electrical Engineering, Control systems, Dynamic systems modeling, Vehicle powertrain optimization and Electric vehicle technology. He is currently a graduate research assistant and has worked on the projects involving Battery pack development, electric vehicle design optimization and powertrain modeling. He also has more than five years of industry experience in the aftersales and diagnostics of Electric vehicles and fast charging stations.

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